

Gaokerena: A Small Persian Medical Language Model Family

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Abstract

The integration of artificial intelligence into medical question-answering systems has advanced rapidly; however, research remains predominantly focused on English, leaving low-resource languages like Persian significantly underserved. To address this gap, this paper introduces Gaokerena, a novel family of compact Persian medical language models optimized for deployment on consumer-grade hardware. As a foundational step toward localized digital healthcare, we first present Gaokerena-V, developed by training a baseline model on a newly curated 90-million-token Persian medical corpus and 20,000 expert-vetted physician Q&A pairs, which improved performance on a translated medical MMLU benchmark from 46.28% to 49.31%. Second, recognizing the critical demands of clinical reasoning, we developed Gaokerena-R by integrating a Chain-of-Thought approach with two novel Reinforcement Learning with AI Feedback (RLAIF) frameworks to optimize preference-based reasoning. Despite utilizing the same baseline architecture and a smaller dataset than Gaokerena-V, Gaokerena-R achieved a superior benchmark score of 52.98%. Furthermore, both models are equipped with custom-developed uncertainty heads that predict the model’s confidence in its responses based solely on internal hidden states. While these results demonstrate significant progress in Persian medical

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language modeling and proactive safety estimation, current performance levels remain insufficient for direct clinical application, highlighting the necessity for further research into robust knowledge acquisition and rigorous safety verification prior to real-world deployment.

Keywords: small medical language model, low resource languages, reasoning language models, reinforcement learning with ai feedback

1. Introduction

The advent of the transformer architecture, introduced in the seminal work “Attention is All You Need” [?], has catalyzed a rapid evolution in the field of natural language processing (NLP). This architectural breakthrough has enabled the development of increasingly sophisticated Large Language Models (LLMs) that leverage self-attention mechanisms to comprehend and generate human language with unprecedented accuracy. Consequently, the integration of artificial intelligence (AI) has surged across various high-stakes domains. This is particularly evident in the medical field, where AI-driven solutions are actively deployed to enhance diagnostic precision, optimize patient care, and streamline administrative workflows.

Modern language models span a vast spectrum of scales, ranging from lightweight models with a few billion parameters to massive architectures exceeding trillions of parameters. While ultra-large models exhibit remarkable general knowledge and highly accurate reasoning, their immense computational requirements preclude deployment on consumer-grade, local hardware. In clinical and medical contexts, this operational bottleneck introduces significant privacy and data security concerns, as sensitive patient information must often be transmitted to external cloud servers. Accordingly, there is a critical and growing imperative for specialized, small medical language models capable of local execution on edge devices to ensure strict data confidentiality.

Despite the extensive research and development dedicated to English-centric small medical language models, such as Medreason [?], a profound disparity

remains regarding the resources available for non-English languages. This resource scarcity is highly pronounced for the Persian language; despite its large, global community of speakers, the complete absence of a high-performing, localized small medical language model leaves a critical gap in clinical AI accessibility. To the best of our knowledge, at the beginning of this research, the only existing Persian medical language model was Sina-BERT [?], which is a closed-source solution with respect to both weights and training data. Furthermore, Sina-BERT is restricted to a non-generative encoder architecture, severely limiting its adaptability for conversational clinical tasks. This lack of open-source, generative alternatives severely hinders reproducibility and slows down the integration of AI solutions within the Persian-speaking medical community.

Beyond the absence of localized, small medical language models, the Persian medical NLP landscape suffers acutely from data scarcity; paradoxically, this gap stems from the underutilization of existing digital assets rather than a fundamental lack of raw source material. Active Persian-language medical forums (e.g., Drhast, Doctor-yab) and authoritative online medical magazines (e.g., Hidarcor, Niniban) host vast repositories of expert-curated content and authentic patient-doctor interactions that, if systematically crawled, rigorously cleaned, and properly structured, represent invaluable corpora for domain-specific language modeling. Furthermore, high-stakes medical examinations administered annually present an untapped resource for academic evaluation and fine-tuning if systematically extracted from legacy PDF formats. Complementing these localized efforts, the strategic translation of established, high-quality English medical datasets offers another vital trajectory to alleviate data scarcity and bridge the resource gap in Persian medical AI.

To address these compounding challenges, this paper introduces a comprehensive suite of data resources and open-source language models tailored specifically for the Persian medical domain. Utilizing these diverse datasets, we develop and introduce two localized small medical language models: **Gaokerena-V** and **Gaokerena-R**, the latter of which incorporates advanced reasoning capabilities. Furthermore, to validate their efficacy, we rigorously evaluate the

performance of both models using our newly curated and structured benchmark datasets.¹ The core contributions of this work are summarized as follows:

- **A 90-Million Token Medical Corpus:** A large-scale, domain-specific text corpus systematically crawled and curated from authoritative Persian medical magazines.
- **The MF3QA Dataset:** A novel, high-quality dataset of 24,000 authentic Persian patient-doctor interaction pairs, curated from active medical forums. Through a meticulous cleaning and filtering process, we distilled this subset from an initial pool of 184,000 raw interaction pairs to ensure clinical relevance and linguistic quality. The dataset is partitioned into 20,000 training, 2,000 development, and 2,000 testing samples.
- **A Consolidated Benchmarking Suite:** A 3,000-question medical MCQ benchmark comprising both official questions from the Iranian Basic Medical Science Exam (IBMSEE) and English-language medical questions expertly translated into Persian to ensure clinical accuracy and linguistic precision.
- **Translated Training Resources:** A curated collection of 18,000 Persian medical multiple-choice questions generated via machine translation of English-language medical sources. These are intended exclusively for training to mitigate the impact of potential translation artifacts on final evaluation metrics.
- **Novel Reasoning Frameworks:** The development of two distinct structural frameworks designed specifically to enhance the medical reasoning capabilities of localized language models, which directly underpinned the training of Gaokerena-R.

¹All research artifacts, including the curated datasets, model weights, and generated model responses, are publicly available at github.com/Mehrdadghassabi/Gaokerena-V and github.com/Mehrdadghassabi/Gaokerena-R.

- **Localized Persian Medical Language Models:** The introduction of Gaokerena-V and Gaokerena-R, two pioneering small Persian medical language models optimized for consumer hardware; while they demonstrate robust medical knowledge retrieval, we explicitly note they are research artifacts and not yet cleared for real-world clinical deployment.
- **Uncertainty Estimation Framework:** The design and implementation of dedicated uncertainty estimation heads for the Gaokerena models. These heads leverage the model’s internal states to quantify predictive uncertainty, enabling confidence scoring and risk mitigation during medical inference.

2. Literature Review

2.1. English Small Medical Language Models

Substantial advancements in specialized medical language models have focused heavily on extracting cognitive processes and structural reasoning. Prominent among these developments is the Meerkat model family [?], which introduces a framework centered on extracting medical Chain-of-Thought (CoT) [?] reasoning trajectories directly from authoritative medical textbooks. By fine-tuning a foundational language model on these reasoning paths alongside complementary biomedical datasets, Meerkat shifts the focus from simple factual retrieval to the underlying decision-making processes inherent to clinical practice. This design allows the model to simulate expert medical reasoning, making it highly effective for complex, multi-step clinical dialogues.

Advanced reasoning has also been pursued through algorithmic search and process evaluation, as demonstrated by the MedSSS [?] framework. To enhance medical reasoning, researchers meticulously structured the intermediate steps between a question and its final answer by utilizing the Monte Carlo Tree Search (MCTS) algorithm over multiple-choice medical question datasets to generate systematic, step-by-step reasoning paths. This process yielded three distinct synthetic datasets for training: a supervised fine-tuning dataset to train

the policy model with high-quality reasoning chains; a preference dataset for Reinforcement Learning (RL) to help the model distinguish between strong and weak responses; and a soft-label dataset to train a process reward model. Ultimately, the trained policy model serves as the primary reasoning engine, while the process reward model acts as a guide during generation, continuously evaluating the logical coherence and medical accuracy of each reasoning step.

While the aforementioned models focus on enhancing reasoning capabilities, MedMobile [?] adopts a different approach to addressing low-resource medical languages. Utilizing Microsoft’s Phi-3-mini [?] as its baseline architecture, MedMobile was fine-tuned on a blended dataset of human-curated and synthetic medical data. This optimization enables the model to run efficiently on resource-constrained mobile devices, providing ubiquitous and privacy-preserving access to high-quality clinical guidance without the need for persistent cloud connectivity.

2.2. Persian Small Language Models

Although many current multilingual models include support for Persian, smaller-scale variants frequently suffer from language mixing, grammatical issues, and a lack of linguistic nuance. This limitation is driven by a severe data imbalance, as these models are trained predominantly on massive English and Chinese corpora, leaving Persian as a negligible fraction of the training data. To bridge this gap, the Cohere for AI research team introduced Aya-Expanse [?], a family of multilingual models featuring a specialized training paradigm. Instead of relying entirely on imbalanced, human-generated web data, Aya-Expanse utilizes high-quality synthetic data generated by larger, state-of-the-art models. While this approach grants the model a sophisticated grasp of Persian syntax and semantics, its general-purpose nature results in performance degradation within highly specialized domains, particularly regarding medical knowledge and clinical reasoning.

Another effort to address native NLP limitations, researchers at the University of Tehran developed PersianMind [?], an open-source bilingual model

optimized for cross-lingual Persian-English tasks. Given the historical shortage of multilingual models that supported Persian when the project began, the developers expanded the tokenizer of LLaMA-2 [?] to support Persian tokens and trained the network to master the language’s syntax and semantics. Yet, much like Aya-Expanse, PersianMind functions strictly as a general-purpose model. Devoid of exposure to deep, domain-specific clinical corpora during its pre-training and instruction-tuning phases, it exhibits the same constraints in advanced clinical reasoning and medical QA. Additionally, Aya-Expanse benefits from a much larger training corpus, granting it superior general knowledge. For these reasons, we ultimately chose Aya-Expanse as our baseline architecture, focusing our efforts on mitigating its specialized domain deficiencies for healthcare applications.

3. Data Collection

As previously noted, the Persian medical domain severely lacks both specialized small language models and high-quality datasets suitable for research. Consequently, to develop and evaluate Persian medical language models, it was essential to curate a proprietary dataset tailored to these research requirements. The data collection pipeline integrated various methodologies, including human and machine translation of existing foreign-language datasets, web scraping diverse sources, and extracting multiple-choice question (MCQ) from PDF documents to build a comprehensive and functional corpus.

3.1. Persian Medical Corpus

The absence of a dedicated Persian medical corpus poses a significant challenge for researchers and developers aiming to build Persian medical models. Without the high-quality, domain-specific text data essential for training artificial intelligence models, these efforts may face major bottlenecks, ultimately hindering the development of advanced medical technologies and solutions tailored for the Persian-speaking population. To address this issue, we have compiled

a comprehensive dataset comprising approximately 90 million tokens across roughly 102,000 articles, each authored by medical professionals.²

Garcia-Ferrero et al. [?] compiled a collection of medical texts spanning four languages (English, French, Spanish, and Italian), which can be compared to our dataset as illustrated in Table ???. The corpus we have gathered was crawled from online medical journals; the specific contribution of each journal to this corpus is displayed in Figure ??.

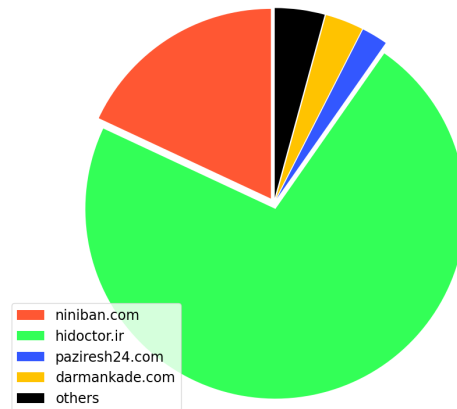


Figure 1: Medical corpus sources

language	no. tokens	collected by
English	1.1B	I. Garcia Ferrero et al.
Spanish	950M	I. Garcia Ferrero et al.
French	675M	I. Garcia Ferrero et al.
Italian	143M	I. Garcia Ferrero et al.
Persian	90M	us

Table 1: Comparison of Our Corpus with Corpora Collected by I. Garcia Ferrero et al.

²To access this corpus, you can visit huggingface.co/datasets/gaokerena/medical_corpus.

3.2. MF3QA

The collection of a real-world doctor-patient question-answering (QA) dataset is crucial for enhancing the capabilities of large language models (LLMs) in the healthcare domain. Despite the existence of reputable online medical forums like Drhast, the Persian language lacked a comprehensive dataset of this nature at the onset of this research; therefore, we have introduced the first Medical Free-Form Farsi Question Answering dataset (MF3QA).³

Such a dataset allows models to learn valuable information derived from authentic interactions between healthcare providers and patients. By analyzing these real-world exchanges, language models can grasp the nuances of medical terminology, patient concerns, and the context surrounding healthcare inquiries. Furthermore, this dataset equips models with the ability to learn not just the factual content of responses but also the appropriate structure and tone for professional communication. This dual learning process is essential, as it enables the model to generate accurate, empathetic, and contextually relevant responses, ultimately improving patient communication and support in medical environments. In this context, Yang Liu [?] highlights several real-world doctor-patient question-answering datasets in his survey; a comparison of these datasets with ours is provided in Table ??.

To construct our dataset, we initially crawled more than 180,000 raw question-answer (QA) pairs from prominent Persian medical forums; the individual contributions of each forum are illustrated in Figure ?. We subsequently implemented a rigorous hybrid pipeline combining automated filtering with manual verification. Our data curation methodology draws inspiration from the approach introduced by Li et al. [?] for the ChatDoctor model, in which approximately half of the scraped English QA pairs were discarded based on answer length, as truncated responses can inhibit a Large Language Model’s (LLM) capacity to generate comprehensive explanations.

However, we encountered a significantly more pronounced data-filtering bot-

³To access this dataset, you can visit huggingface.co/datasets/gaokerena/MF3QA.

Dataset Name	Language	No. Records
ChatDoctor [?]]	English	100K
CMtMedQA [?]]	Chinese	68K
DISC-Med-SFT [?]]	Chinese	465K
HuatuoGPT-sft-data-v1 [?]]	Chinese	226K
Huatuo-26M [?]]	Chinese	26M
MedDialog [?]]	Chinese & English	3.66M
Medical-Meadow [?]]	English	160K
MF3QA	Persian	24K

Table 2: Comparison of our dataset with existing medical QA datasets

tleneck: Persian medical practitioners online tend to provide much more concise responses than their English-speaking counterparts. Consequently, to maintain high training quality and clinical relevance, we filtered out over 80% of our raw records. Furthermore, our preprocessing pipeline employed a more sophisticated multi-stage strategy than the simple length thresholds utilized in ChatDoctor. First, to shift the token-count distribution toward longer, more informative responses, we applied a randomized omission strategy for answers containing fewer than 50 tokens within the training and development sets. To preserve the integrity of our evaluation metrics, this random exclusion was not applied to the test split; instead, we manually reviewed every QA pair in the test set to verify factual accuracy and clinical validity. For the test set specifically, we extracted data from the Doctor-yab and Isovisit platforms and ensured linguistic diversity by translating the K-QA question-answering dataset [?]] and appending it to our test split. Additionally, every record underwent intensive manual review to eliminate spam, duplicates, and low-quality entries. This rigorous process yielded a final dataset consisting of 20,000 QA pairs for the training set, 2,000 for the development set, and 2,000 for the test set.

It is important to note that the Drhast platform does not provide all doctor-patient interaction records directly; it only offers public access to the 2,000 most

recent dynamic records. Furthermore, each record is linked to approximately 100 related entries, significantly complicating the crawling process. To address this challenge, we modeled the data as a graph and performed a Breadth-First Search (BFS) traversal. The BFS execution utilized an iterative loop, selecting available records from the initial 2,000 as root nodes to explore their linked neighbors. By repeatedly performing BFS with new roots, we ensured broader coverage of the dataset while preventing redundancy through post-iteration duplicate filtering. Because the Drhast records are dynamic and change as new questions are posted, we were able to utilize newly added records as roots for subsequent BFS runs. This iterative process successfully extracted 120,000 records out of an estimated 200,000 within a two-week period.



Figure 2: MF3QA sources

3.3. Persian Medical Multiple Choice Question Dataset

While free-form question-answering (QA) datasets are invaluable, the inclusion of multiple-choice question (MCQ) datasets is equally critical for the development of robust medical language models. Various training paradigms, such as those utilized for introducing the MedSSS medical language model [?], specifically require high-quality MCQ datasets to facilitate specialized instruction tuning. Furthermore, evaluating a medical language model using a standardized MCQ test set remains the gold standard for measuring clinical reasoning, as it provides an objective, quantifiable metric for assessing a model’s diagnostic and pharmacological knowledge.

To establish a rigorous benchmark for Persian medical language models, we curated two distinct, high-precision test sets: firstly We have extracted approximately 2,000 questions from the Iranian Basic Medical Science Entrance Exams (2021–2024). These questions were meticulously converted from PDF formats into a structured digital format to serve as a domain-specific, localized benchmark.⁴ secondly We performed a meticulous manual translation of the medical portion of the Massive Multitask Language Understanding (MMLU) dataset [?], comprising approximately 1,000 questions. This allows for standardized cross-lingual comparison with global benchmarks.⁵

Recognizing that training data is less sensitive to minor semantic variations than evaluation data, we employed a scalable machine translation framework to develop a large-scale training corpus. Given the absence of a comprehensive Persian medical MCQ dataset at the time of this research, we utilized the cost-efficient yet highly capable DeepSeek-V3 model [?] to translate a significant portion of the MedMCQA dataset [?] from English to Persian.⁶

To ensure thematic diversity, questions were sampled randomly from the MedMCQA pool, ensuring broad coverage of medical disciplines, including physiology, pharmacology, pathology, and clinical practice. To maintain high data quality, we implemented a dual-referee validation process using two independent models: Grok-3-mini [?] and GPT-4.1-mini [?]. Each referee was tasked with evaluating the translated questions and their corresponding options for semantic accuracy, linguistic fluency, and the preservation of specialized medical terminology. A translation was only accepted into our final dataset if both referees independently assigned it a perfect score of five out of five. Through this rigorous, multi-stage process, we successfully curated a high-quality dataset consisting of approximately 18,000 medically accurate, Persian-language multiple-choice questions.

⁴To access this dataset, you can visit huggingface.co/datasets/gaokerena/KOPP.

⁵To access this dataset, you can visit huggingface.co/datasets/gaokerena/FA_MED_MMLU.

⁶To access this dataset, you can visit github.com/Mehrdadghassabi/Gaokerena-R/tree/main/dataset/medmcqa_translation_filterd.

4. Gaokerena Models

We introduce two lightweight Persian medical language models, Gaokerena-V and Gaokerena-R, both utilizing Aya-Expanse-8B as their foundational architecture. The former, Gaokerena-V, is developed via domain adaptation, trained on a strategically selected subset of our curated medical corpus in conjunction with the MF3QA dataset to optimize for medical proficiency and efficiency. The latter, Gaokerena-R, is designed to enhance logical reasoning through specialized post-training on Multiple-Choice Questions (MCQs). By utilizing our proposed framework to distill reasoning from a larger teacher model, Gaokerena-R is specifically optimized for medical reasoning.

4.1. Gaokerena-V

To effectively adapt the 8-billion-parameter *Aya-Expanse* checkpoint to the nuances of the Persian medical domain, we adopted a structured two-stage fine-tuning strategy consisting of a domain-adaptation stage and an instruction-tuning stage. This two-stage paradigm establishes a coherent training pipeline: the initial stage facilitates the internalization of dense medical knowledge and factual information, providing a robust foundation for the subsequent stage, which focuses on refining instruction-following capabilities and the communication protocols required for effective medical QA. Across both stages, the baseline model was trained on a total of 60 million new tokens. A comprehensive summary of the hyperparameters utilized for both stages is provided in Table ??.

During the domain-adaptation stage, we performed domain-adaptive pre-training on 60% of our curated medical corpus. This phase was designed to immerse the model in specialized Persian medical terminology, linguistic structures, and foundational clinical concepts. To ensure computational efficiency and mitigate the risk of catastrophic forgetting of the model’s general-purpose capabilities, we employed Low-Rank Adaptation (LoRA) [?]. By injecting trainable rank-decomposition matrices into the self-attention layers, we captured domain-specific nuances without requiring a full-parameter update. The

steady convergence of the training process, reflecting the successful internalization of the medical corpus, is illustrated by the decreasing loss trajectory in Figure ??.

In the subsequent instruction-tuning stage, we transitioned to specialized tuning aimed at refining the model’s conversational fluency. We utilized the full training split of our newly constructed MF3QA dataset, which comprises high-quality, real-world doctor-patient interactions. To further optimize the model for task-specific performance specifically the interpretation of symptoms and the provision of contextually relevant medical guidance we continued using the LoRA framework. To prevent overfitting during this stage, we employed more conservative hyperparameters, specifically utilizing a lower LoRA rank and alpha, alongside an increased LoRA dropout rate. The loss trajectory for this instruction-tuning phase is illustrated in Figure ??.

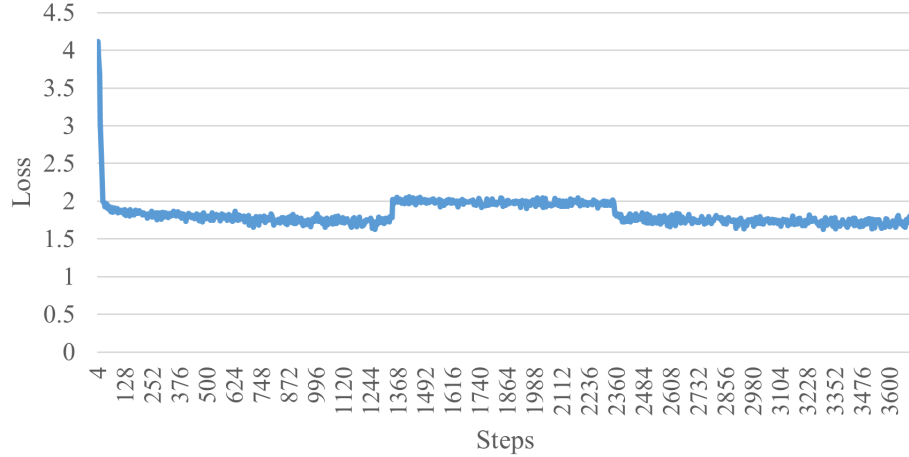


Figure 3: fine-tuning loss curve

4.2. Gaokerena-R

To introduce **Gaokerena-R**, which enhances the reasoning capabilities of the Aya-expanse-8b baseline, we propose a two-pronged Reinforcement Learning from AI Feedback (RLAIF) [?] pipeline. Our objective is to transform the base model into a specialized medical reasoning agent by constructing a high-quality

	Domain-adaptation	Instruction-tuning
on	medical_corpus	MF3QA
Number of epochs	1	1
Batch size	2	2
Number of gradient accumulation steps	16	16
Optimizer	AdamW	AdamW
Learning rate	5E-4	5E-4
Maximum gradient norm	0.3	0.3
Warmup ratio	0.03	0.03
Weight decay rate	0.1	0.5
Maximum context length	1024	1024
Padding strategy	left side padding	left side padding
Lora rank	8	2
Lora alpha	16	2
Lora dropout rate	0.05	0.4
Target modules	all linear layers	all linear layers

Table 3: Gaokerena-V Training Hyperparameters

preference dataset $\mathcal{D} = \{(x, y_w, y_l)\}$, where x denotes a medical question, y_w is a preferred reasoning trajectory, and y_l is a rejected response. We then optimize the student policy π_θ using Direct Preference Optimization (DPO) [?] to align it with these high-quality trajectories. The optimization objective is defined by the DPO loss in Formula ??, and its convergence is illustrated by the DPO loss curve shown in Figure ??.

To enhance the medical reasoning capabilities of the student policy π_θ , we employ a teacher-student architecture where the student (π_S) is the baseline **Aya-Expanse-8b** model and the teacher (π_T) is a high-capacity reasoning model (DeepSeek-R [?]). We generate preference pairs using two complementary strategies to provide diverse training signals; these signals allow the student to

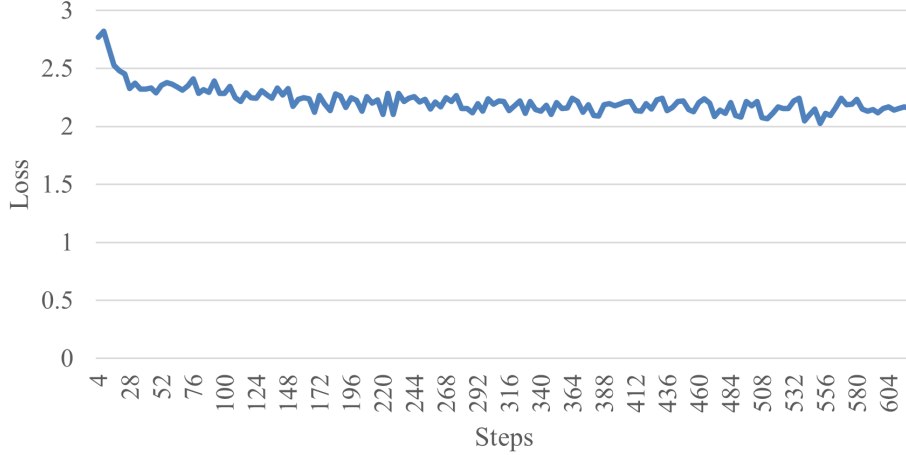


Figure 4: instruction-tuning loss curve

learn medical reasoning behaviors from the teacher.

The first strategy, *Expert-Guided Synthesis*, contrasts the baseline model’s flawed logic with expert-level reasoning. For a given question x , the student model π_S produces an initial response y_{init} . If y_{init} is deemed incorrect, it is assigned as the rejected response (y_l). To obtain the preferred trajectory (y_w), the teacher model π_T is prompted with the ground-truth answer and instructed to generate a detailed, correct Chain-of-Thought (CoT) explanation. This yields a preference triple (x, y_w, y_l) that explicitly separates erroneous student reasoning from verified expert trajectories, as illustrated in Figure ??.

The second strategy, *Iterative Self-Correction*, encourages intrinsic error correction within the student’s own distribution via an iterative feedback loop. To remain computationally efficient while maximizing the utility of each sample, we use a critique-based refinement process. If y_{init} is incorrect, the teacher π_T generates a textual critique c that highlights the conceptual error without revealing the correct solution. The student is then prompted with this critique to produce a corrected response y_{retry} . If y_{retry} is correct, it becomes the preferred response (y_w), while the original incorrect response y_{init} serves as the rejected response (y_l). This design grounds preferred trajectories in the student’s latent

space, promoting more stable convergence. The overall workflow is depicted in Figure ??.

During data synthesis, we find that *Expert-Guided Synthesis* is the dominant contributor to the preference dataset construction; for every 19 preference pairs generated by this strategy, typically only one pair is successfully synthesized through *Iterative Self-Correction*. This hybrid scheme balances expert-led guidance with autonomous student correction. The final preference dataset contains approximately 11,000 pairs, totaling roughly 2 million preferred tokens and 2.5 million rejected tokens.

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right] \quad (1)$$

where σ denotes the sigmoid function and β controls the strength of the update relative to the reference model. Minimizing $\mathcal{L}_{\text{DPO}}$ implicitly increases the probability of valid reasoning trajectories (y_w) while suppressing flawed logic (y_l), thereby distilling the teacher’s reasoning into the Gaokerena-R model.

5. Uncertainty Heads

Our methodology builds upon the LLM Uncertainty Head framework introduced in [?], which augments a pretrained language model with a lightweight token-level classification head to detect hallucinated claims during generation. The core idea is to train this auxiliary head to predict, for each output token, whether the corresponding claim is factually supported or fabricated, thereby providing an internal uncertainty signal without relying on external retrieval or sampling-based methods. This design is particularly appealing because it adds minimal parameter overhead and can be trained efficiently alongside the frozen base model.

In its original implementation for the Mistral-7b model [?] on English-language data, the framework follows a clear pipeline: it extracts atomic, verifiable claims from the model’s generated responses, aligns these claims with

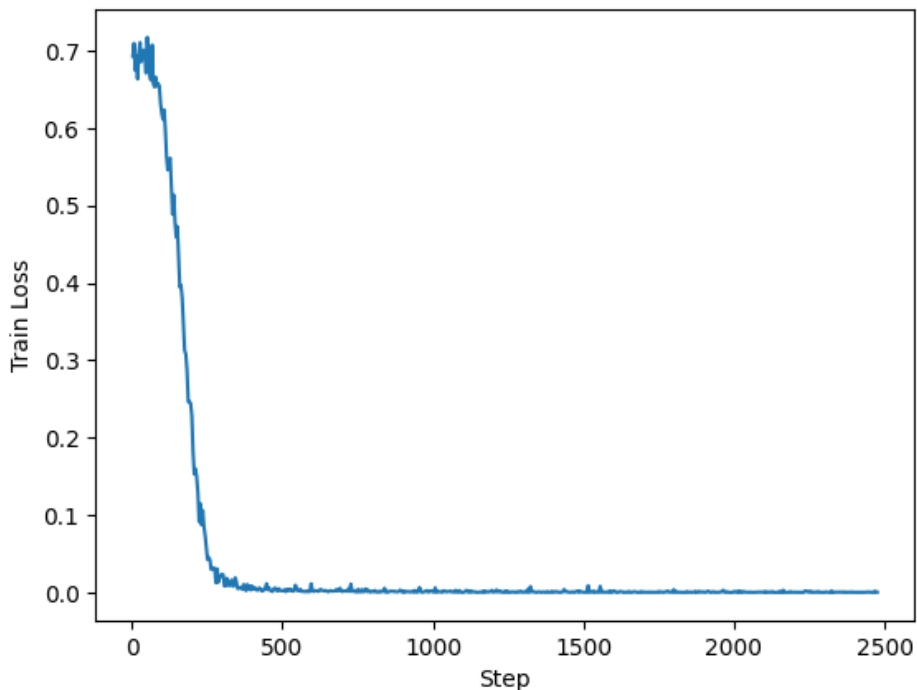


Figure 5: DPO loss curve

their exact token spans, and trains the classification head using ternary labels 1 for hallucinated tokens, 0 for supported ones, and -100 to mask out all non-claim tokens (e.g., stopwords, punctuation, and auxiliary phrases) so they do not contribute to the loss. This design ensures that the head learns exclusively from semantically meaningful content, which is critical for accurate uncertainty estimation.

In this work, we extend this paradigm to the gaokerena family of models to support Persian. This transfer, however, introduces several non-trivial challenges. Persian is written in a right-to-left script and exhibits rich morphological inflections, which complicate straightforward substring matching and span extraction. Moreover, the gaokerena tokenizer operates with a distinct subword vocabulary and byte-pair encoding merges compared to Mistral, meaning that a single claim token in English may correspond to a sequence of multiple subword tokens in Persian. To overcome these issues, we design a tailored data

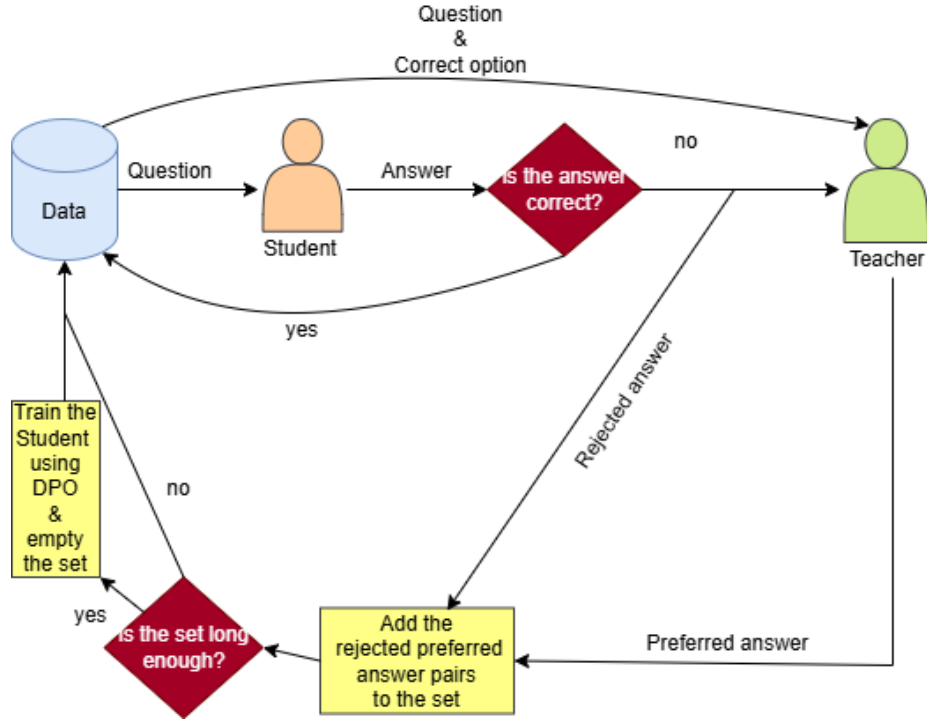


Figure 6: Method 1 Block Diagram

construction pipeline that carefully preserves both semantic meaning and structural consistency across languages. This involves translating and re-validating each atomic claim to ensure it remains a self-contained, verifiable unit in Persian before any annotation takes place. As a result, we construct a new uncertainty dataset specifically curated for Persian, containing grounded claim annotations that are directly usable for our task.

Finally, we perform a fine-grained token-level alignment on the Persian text, where we map each claim token to its precise position in the Persian sentence, explicitly accounting for subword splits and the tokenizer’s idiosyncrasies. We assign the appropriate binary labels to the identified spans and mask all remaining tokens with -100 to exclude them from the loss. Throughout this process, we keep the base gaoakerena model frozen and only update the uncertainty head, preserving the model’s original generative capabilities while learning a reliable

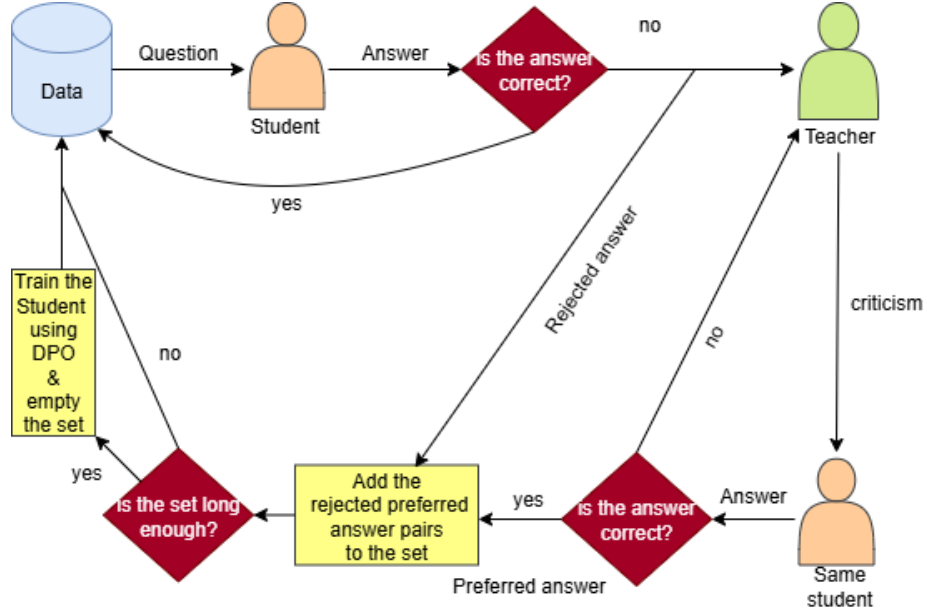


Figure 7: Method 2 Block Diagram

uncertainty signal. This yields a training set that is fully compatible with the gaokerena tokenizer’s vocabulary and seamlessly integrates with the uncertainty head’s learning objective, enabling effective cross-lingual transfer while maintaining the core strengths of the original framework.

6. Results

As previously discussed, multilingual Small Language Models (SLMs) often struggle with underrepresented languages, including Persian. Cohere For AI addressed this challenge by introducing Aya Expanse, which demonstrates significant improvements in Persian fluency and linguistic comprehension. However, the curse of multilinguality persists, often resulting in performance trade-offs. In this research, we aim to mitigate these limitations by exploring various methods to enhance medical reasoning capabilities within the Persian context. Consequently, this section presents a comprehensive evaluation of our proposed models, benchmarking their performance against established alternatives to as-

sess their effectiveness in specialized medical tasks.

6.1. *Gaokerena-V Results*

Given the scarcity of publicly available, specialized Persian medical Small Language Models (SLMs), we benchmarked the Gaokerena-V model against several general-purpose alternatives, including its baseline (Aya-Expanse-8B), PersianMind, and Qwen2.5 [?].

As presented in Table ??, Gaokerena-V achieved a significant milestone by surpassing the 36% passing threshold of the Iranian Basic Medical Sciences Entrance Exam. To our knowledge, this makes it the first Persian-language model under 8 billion parameters to successfully pass this examination. Furthermore, Gaokerena-V demonstrated superior performance on the translated MMLU dataset, consistently outperforming other models across most sub-categories, thereby validating its efficacy in medical knowledge synthesis and generation in Persian.

Notably, the model achieved a 7.4% improvement in the anatomy category over the baseline. We attribute this gain to the relative abundance of anatomy-focused material in reputable Persian digital publications. Conversely, the more moderate gains in other sub-categories highlight a critical bottleneck: the lack of comprehensive, high-quality medical corpora in Persian, which restricts further performance optimization.

Finally, Table ?? reveals that while PersianMind exhibits low inference latency, this is primarily an artifact of its tendency to produce truncated, overly concise responses. In several categories, PersianMind’s performance proved negligible, effectively mirroring the scores of random chance, which underscores its limitations in complex medical reasoning tasks.

6.2. *Gaokerena-R Results*

As a medical-reasoning-enhanced version of Aya-Expanse-8b, Gaokerena-R is designed to leverage structured reasoning paths. However, our experimental results show that its superior performance is only unlocked when utilizing a Chain

	gao kerena-V (ours)	Aya- Expanse-8b (baseline)	Qwen 2.5	Persian Mind
MMLU-anatomy(fa)	48.14	40.74	41.48	25.18
MMLU- medical-genetics(fa)	53.0	49.0	52.0	34.0
MMLU- college-medicine(fa)	43.93	44.51	43.35	20.23
MMLU- clinical-knowledge(fa)	55.47	52.07	47.92	25.28
MMLU- professional-medicine(fa)	47.05	45.58	43.01	23.89
MMLU-college-biology(fa)	47.22	45.14	42.36	32.63
MMLU(avg)	49.31	46.64	45.17	25.89
IBMSEE Sept2023	38.69	34.52	33.33	19.64
Number of parameters	8b	8b	7.6b	6.8b
Inference time	$\approx 10s$	$\approx 10s$	$\approx 15s$	$\approx 2s$

Table 4: Gaokerena-V Performance Comparison with Other models

of Thought (CoT) prompting strategy. Without CoT, the model fails to outperform its baseline backbone, as demonstrated by the direct (straight) prompting comparison with Gaokerena-V and Aya-Expanse-8b in Table ?? . Therefore, to achieve optimal performance with Gaokerena-R, it is essential to configure the system prompt to explicitly instruct the model to reason step-by-step.

To evaluate Gaokerena-R against the baseline models, we generate five responses per question and apply a modified self-consistency (CoT-SC) decoding strategy [?]. A majority agreement threshold of at least three out of five runs (3/5) is required to select the final answer; when no single option reaches this threshold, Aya-Expanse-8B serves as an external verifier to break the tie and determine the final selection. For the baseline comparison, Gaokerena-V

	Gaokerena-R	Gaokerena-V	Aya-Expanse-8b -8b (backbone)
MMLU-anatomy(fa)	41.48	48.14	40.74
MMLU- medical-genetics(fa)	49.0	53.0	49.0
MMLU-college-medicine	46.24	43.93	44.51
MMLU- clinical-knowledge	52.45	55.47	52.07
MMLU- professional-medicine	41.91	47.05	45.58
MMLU-college-biology	44.44	47.22	45.14
MMLU(avg)	46.28	49.31	46.64
IBMSEE Sept2023	35.11	38.69	34.52
Number of parameters	8b	8b	8b
Prompt type	Straight	Straight	Straight
Inference time	$\approx 10s$	$\approx 10s$	$\approx 10s$

Table 5: Performance of Gaokerena models with Straight Prompt

and Aya-Expanse-8B are evaluated using standard direct prompting (without self-consistency) to establish a fair reference. As reported in Table 6, while Gaokerena-R incurs higher inference latency due to its multi-run decoding process, it delivers substantial performance gains—outperforming Gaokerena-V by 3.67 percentage points and Aya-Expanse-8B by 6.34 percentage points on the Persian-translated medical subset of the MMLU dataset.

To account for the high response diversity observed in Gaokerena-R, we adopted the metric proposed by Brown et al. [?], which is designed to explore response variability through the *pass@k* metric—the probability that at least one correct answer is produced within *k* attempts. We evaluated our models by generating five samples per question and reporting the averaged *pass@k* for

	Gaokerena-R	Gaokerena-V	Aya- Expense-8b
MMLU-anatomy(fa)	47.40	48.14	40.74
MMLU-genetics(fa)	56.0	53.0	49.0
MMLU-college-medicine	50.28	43.93	44.51
MMLU-clinical-knowledge	58.86	55.47	52.07
MMLU-professional-medicine	48.89	47.05	45.58
MMLU-college-biology	54.86	47.22	45.14
MMLU(avg)	52.98	49.31	46.64
IBMSEE Sept2023	46.42	38.69	34.52
Prompt	COT	Direct	Direct
Decoding	COT_SC (with verifier)	Zero-shot	Zero-shot
Inference time	$\approx 5 \times 35 + 10 + 8s$	$\approx 10s$	$\approx 10s$

Table 6: Performance of Gaokerena models with Preferred Prompting Strategy

$k \leq 3$ across the entire test set (Eq. ??). We omitted higher values of k because our questions are multiple-choice with only four options; thus, a model selecting all options would trivially achieve a perfect score at $k = 4$. As illustrated in Figures ?? and ??, the Gaokerena-R curve consistently outperforms both Gaokerena-V and the Aya-Expense-8b baseline. This trend underscores the superior medical reasoning capabilities acquired by Gaokerena-R through our proposed framework. This advantage is evident across nearly all categories, with the sole exception of Anatomy, where Gaokerena-V performs slightly better; we

attribute this to the fact that anatomy primarily requires factual knowledge retrieval rather than complex reasoning. Furthermore, while Gaokerena-V’s high uncertainty and stochastic response patterns inflate its scores at higher k values (where random guessing becomes more effective), Gaokerena-R’s superior performance at low k demonstrates genuine reasoning proficiency and calibrated confidence.

$$\text{pass@k} = 1 - \frac{\binom{M-C}{k}}{\binom{M}{k}} \quad (2)$$

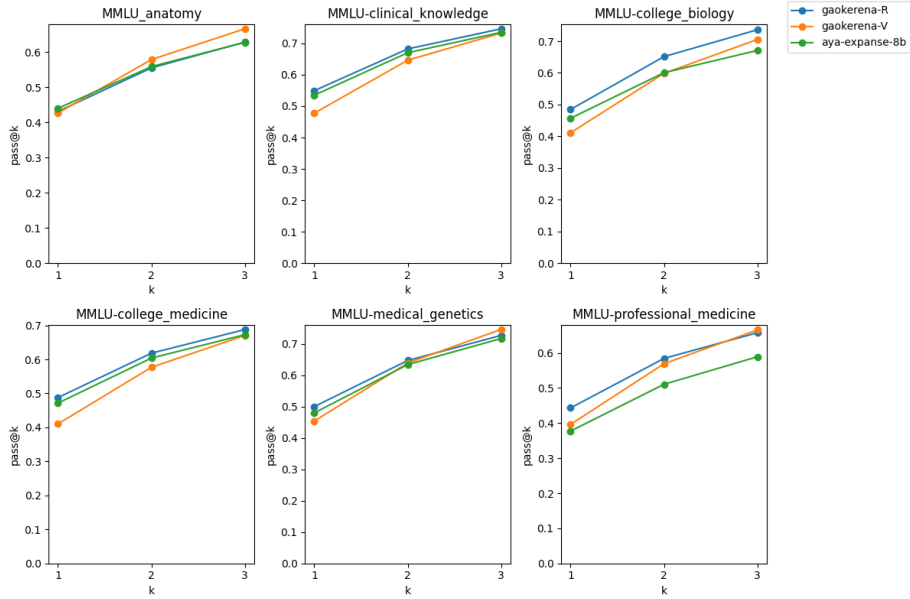


Figure 8: Pass@k results on the Persian Translated Medical Portion of MMLU dataset

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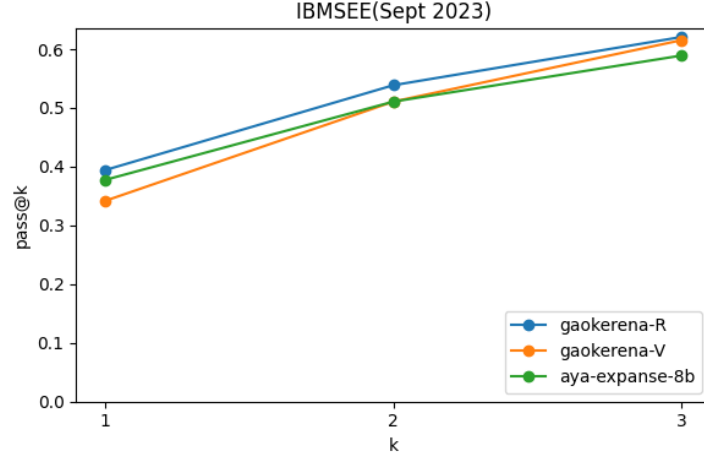


Figure 9: Pass@k results on the IBMSEE Sept2023 dataset

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